

Solution exercise set 1

Problem 1:

a) For an unbiased coin we have $P(H) = P(T) = 0.5$.

$$\begin{aligned}
 P(HHTHT) &= \underbrace{P(H \cap H \cap T \cap H \cap T)}_{\text{due to independent events}} = P(H)P(H)P(T)P(H)P(T) \\
 &= 0.5 \cdot 0.5 \cdot 0.5 \cdot 0.5 \cdot 0.5 = (0.5)^5 = \underline{\underline{0.03125}} \\
 P(THHHT) &= \dots = (0.5)^5 = \underline{\underline{0.03125}}
 \end{aligned}$$

b) See the accompanying R-code solution file.

c) For the exact calculation we observe that this is a situation with a specified number ($n = 5$) of independent trial (independent flipping) with two outcomes (H or T) with the same probability in each trial (0.5 for H in each flip). I.e. the number of successes (heads) has a binomial distribution with $n = 5$ and $p = 0.5$. Then:

$$P(\text{three heads}) = \binom{5}{3} 0.5^3 (1 - 0.5)^{5-3} = 10 \cdot 0.03125 = \underline{\underline{0.313}}$$

d) See R-code.

e) For the exact calculation, the only change is that now the probability of H is 0.6, then:

$$P(\text{three heads}) = \binom{5}{3} 0.6^3 (1 - 0.6)^{5-3} = 10 \cdot 0.03456 = \underline{\underline{0.346}}$$

Problem 2:

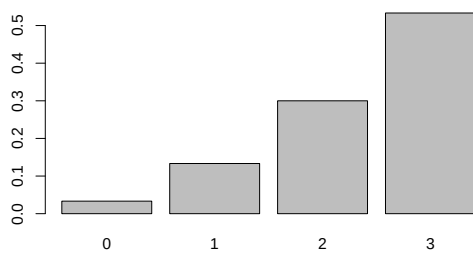
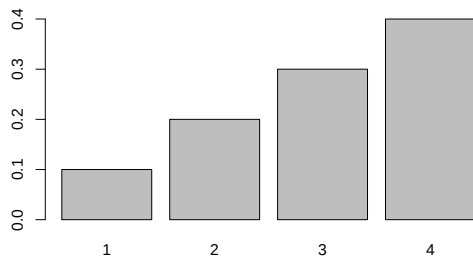
a) Using that $\sum_x f(x) = 1$ we get:

$$\sum_{x=1}^4 f(x) = 1/c + 2/c + 3/c + 4/c = 10/c = 1 \Rightarrow \underline{\underline{c = 10}}$$

b)

$$\sum_{x=0}^3 f(x) = c(0+1)^2 + c(1+1)^2 + c(2+1)^2 + c(3+1)^2 = c \cdot 30 = 1 \Rightarrow \underline{\underline{c = 1/30}}$$

c)



Problem 3:

	2a)	2b)
$E(X) = \sum_x x \cdot f(x)$	3	2.33
$E(X^3) = \sum_x x^3 \cdot f(x)$	35.4	16.93

Problem 4:

$$f(x) = \begin{cases} 4x(1-x^2) & \text{for } 0 \leq x \leq 1 \\ 0 & \text{otherwise} \end{cases}$$

a)

$$P(X < 0.5) = \int_0^{0.5} 4(x - x^3)dx = 4\left[\frac{1}{2}x^2 - \frac{1}{4}x^4\right]_0^{0.5} = 4\left(\frac{1}{2}0.5^2 - \frac{1}{4}0.5^4\right) = \frac{7}{16} = \underline{\underline{0.4375}}$$

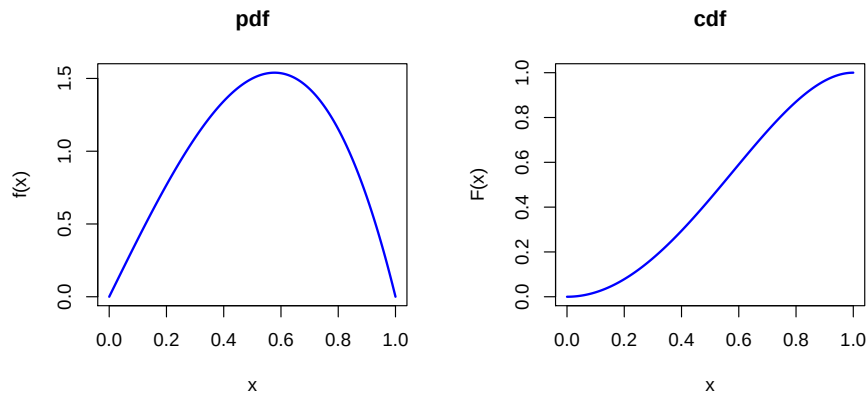
$$\begin{aligned} E(X) &= \int_{-\infty}^{\infty} xf(x)dx = \int_0^1 x4(x - x^3)dx = 4 \int_0^1 (x^2 - x^4)dx \\ &= 4\left[\frac{1}{3}x^3 - \frac{1}{5}x^5\right]_0^1 = 4\left[\frac{1}{3} - \frac{1}{5}\right] = 4 \cdot \frac{2}{15} = \frac{8}{15} = \underline{\underline{0.533}} \end{aligned}$$

$$\begin{aligned}
\text{Var}(X) &= E(X^2) - [E(X)]^2 \\
E(X^2) &= \int_{-\infty}^{\infty} x^2 f(x) dx = \int_0^1 x^2 4(x - x^3) dx = 4 \int_0^1 (x^3 - x^5) dx \\
&= 4 \left[\frac{1}{4} x^4 - \frac{1}{6} x^6 \right]_0^1 = 4 \left[\frac{1}{4} - \frac{1}{6} \right] = 4 \cdot \frac{1}{12} = \frac{1}{3} \\
\text{Var}(X) &= \frac{1}{3} - \left(\frac{8}{15} \right)^2 = \underline{\underline{0.049}} \\
\text{SD}(X) &= \sqrt{\text{Var}(x)} = \sqrt{0.049} = \underline{\underline{0.22}}
\end{aligned}$$

b)

$$\begin{aligned}
F(x) &= P(X \leq x) = \int_{-\infty}^x f(t) dt = \int_0^x 4t(1 - t^2) dt \\
&= \int_0^x (4t - 4t^3) dt = [2t^2 - t^4]_0^x = \underline{\underline{2x^2 - x^4}} \\
P(X < 0.3) &= F(0.3) = 2 \cdot 0.3^2 - 0.3^4 = \underline{\underline{0.1719}} \\
P(X > 0.7) &= 1 - P(X < 0.7) = 1 - F(0.7) = 1 - (2 \cdot 0.7^2 - 0.7^4) = \underline{\underline{0.2601}}
\end{aligned}$$

c)



Problem 5:

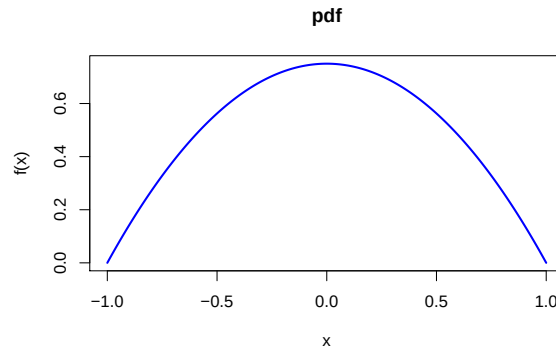
$$f(x) = \begin{cases} k(1 - x^2) & \text{for } -1 \leq x \leq 1 \\ 0 & \text{otherwise} \end{cases}$$

a) For $f(x)$ to be a pdf we must have $\int_{-\infty}^{\infty} f(x) dx = \int_{-1}^1 f(x) dx = 1$.

$$\int_{-1}^1 f(x) dx = k \left[x - \frac{1}{3} x^3 \right]_{-1}^1 = k \left(1 - \frac{1}{3} - (-1 + \frac{1}{3}) \right) = k \frac{4}{3} = 1$$

This gives $\underline{\underline{k = \frac{3}{4}}}$.

b) A sketch of $f(x)$ is given in the figure below.



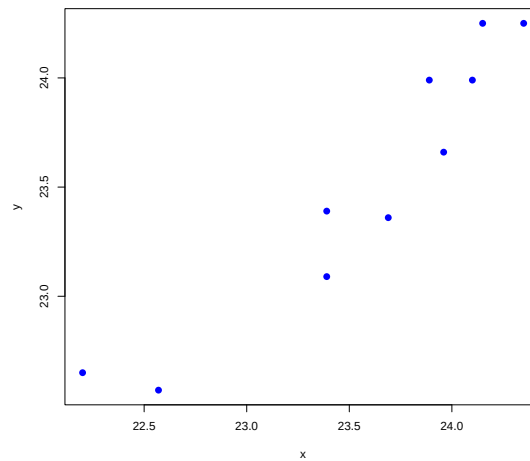
c)

$$P(X \leq 0.5) = \int_{-1}^{0.5} \frac{3}{4}(1-x^2)dx = \frac{3}{4}\left[x - \frac{1}{3}x^3\right]_{-1}^{0.5} = \frac{3}{4}(0.5 - 0.04167 - (-1 + 0.3333)) = \underline{\underline{0.844}}$$

$$\begin{aligned} \text{d)} \quad E(Y) &= E(1 + X^2) = \int_{-\infty}^{\infty} (1 + x^2)f(x)dx = \int_{-1}^1 (1 + x^2)\frac{3}{4}(1 - x^2)dx \\ &= \frac{3}{4} \int_{-1}^1 (1 - x^4)dx = \frac{3}{4}\left[x - \frac{1}{5}x^5\right]_{-1}^1 = \frac{3}{4}\left[2 - \frac{2}{5}\right] = \underline{\underline{1.2}} \end{aligned}$$

Problem 6:

a)



b) The empirical correlations is $r = 0.94$. A strong positive correlation - the gas prices follow each other to a large degree.

Problem 7:

a) If Pepsi and Cola taste the same, the probability of a correct identification is $1/5=0.2$. This goes for all test persons, and if we assume that the persons identify correctly independently of each other, we can conclude that X has the binomial distribution. This is

because we then have a situation with a specified number of trials $n = 10$, the trials are independent, there are two outcomes in each trial and where the probability of successful outcome is the same in each trial, $p = 0.2$. I.e. $X \sim \text{Bin}(n, p)$, $n = 10$ and $p = 0.2$.

$$\begin{aligned} P(X \geq 3) &= 1 - P(X \leq 2) = 1 - \{P(X = 0) + P(X = 1) + P(X = 2)\} \\ &= 1 - \left\{ \binom{10}{0} 0.2^0 0.8^{10} + \binom{10}{1} 0.2^1 0.8^9 + \binom{10}{2} 0.2^2 0.8^8 \right\} = \underline{\underline{0.322}} \\ E(X) &= np = 10 \cdot 0.2 = \underline{\underline{2}} \\ \text{Var}(X) &= np(1 - p) = 10 \cdot 0.2 \cdot 0.8 = \underline{\underline{1.6}} \end{aligned}$$

b) Now we have a situation where independent binary trials with the same probability for success are repeated until first success. I.e. N has a geometric distribution with probability of success $p = 0.2$. We then have that $E(N) = \frac{1}{p} = \underline{\underline{5}}$.

Problem 8:

See the accompanying R-code solution file.

The results are: **a)** 0.221 **b)** 0.062 **c)** 0.438 **d)** 195.1

Problem 9:

a) All four investments have equal expected returns:

$$E(2X) = 2E(X) = 2 \cdot 0.4 = 0.8.$$

$$E(X + Y) = E(X) + E(Y) = 0.4 + 0.4 = 0.8.$$

$$E(X + Z) = E(X) + E(Z) = 0.4 + 0.4 = 0.8.$$

$$E(Y + Z) = E(Y) + E(Z) = 0.4 + 0.4 = 0.8.$$

$$\text{Var}(2X) = 2^2 \text{Var}(X) = 4.$$

$$\text{Var}(X + Y) = \text{Var}(X) + \text{Var}(Y) + 2\text{Cov}(X, Y) = 1 + 1 + 2 \cdot 0 = 2.$$

$$\text{Var}(X + Z) = \text{Var}(X) + \text{Var}(Z) + 2\text{Cov}(X, Z) = 1 + 1 + 2 \cdot 0.5 = 3$$

(covariance equals correlation since all variances are 1).

$$\text{Var}(Y + Z) = \text{Var}(Y) + \text{Var}(Z) + 2\text{Cov}(Y, Z) = 1 + 1 + 2 \cdot (-0.5) = 1.$$

Investment $Y + Z$ is safest since it has the lowest risk (variance) among investments with equal expected returns.